

FINAL CLIENT DEMONSTRATION

FundInv

*Four-role presentation operating guide***The story in one sentence**

FundInv separates access control, investor requests, fund valuation, and financial settlement across four roles so that no single user controls the entire transaction lifecycle.

Audience

OKC, assessors, and non-technical stakeholders

Recommended demo

12–15 minutes

Live portal

<https://dxnh4gknox8f9.cloudfront.net>

Validation record

Production-tested on 4 August 2026

Use this document as the presenter’s script and safety checklist. It records what was actually verified, what remains externally blocked, and what each control means.

1. Readiness verdict

What is safe to demonstrate tomorrow

Core workflow: demonstration-ready

Admin access, Investor fund browsing, demo PayNow subscription, Operations settlement, Manager manual valuation with preview, Investor P&L allocation, redemption, holdings, analytics, audit records, and AWS-hosted access were exercised successfully.

Area	Result	Evidence
Backend test suite	PASS	34/34 unit and integration tests
Frontend production build	PASS	36 routes compiled and prerendered
Live subscription	PASS	\$12.34 matched, settled once
Live redemption	PASS	\$7.89 approved, transferred, completed
P&L allocation	PASS	\$10 fund P&L allocated \$4.00 to a 40% holder
Investor dashboard consistency	PASS	Value changed by net cash flow of +\$4.45
Automatic market suggestion	BLOCKED	AWS runtime receives Alpaca 401 Unauthorized
CI/CD credentials	ATTENTION	GitHub deployment jobs fail at AWS credential setup; manual rollout used

Do not claim zero defects

The controlled manual valuation path works, but automatic Alpaca-derived P&L is not currently available in AWS. Present it as an external-credential limitation, not as a working feature.

2. The four-role control model

Who does what, and why

Role	Owns	Must not do	Hands off to
Admin	Invitations, users, role/access status, oversight	Value funds or settle money	Investor / Manager / Operations
Investor	Chooses a fund and submits subscription/redemption requests	Issue units, set NAV, self-approve	Operations
Manager	Reviews fund performance, previews and finalizes NAV/P&L	Confirm payment or alter investor requests	Operations and Investor reporting
Operations	Verifies money movement and settles units	Change P&L, NAV, or user roles	Investor reporting and Admin audit

End-to-end sequence

1. Admin grants the correct role and leaves an auditable access record.
2. Investor selects a fund, account, and exact amount; demo PayNow records the same amount and reference.
3. Manager finalizes the applicable daily fund valuation so a valid NAV exists.
4. Operations compares request, receipt, fund, status, and valuation before settlement.
5. The database issues or redeems internal fund units exactly once.
6. Investor sees updated holdings, activity, NAV P&L allocation, and analytics; Admin can audit the lifecycle.

Presentation language

Say fund units, subscription, redemption, finalized NAV, and demo PayNow. Do not say the Investor directly owns public QQQ or VOO shares; the Investor owns internal units in a FundInv product.

3. Manager daily valuation — the straight explanation

Yes: the field is intended to prefill automatically

Refresh calculation should obtain Alpaca market prices, calculate a weighted fund return, multiply it by opening assets, and prefill Suggested daily P&L. The Manager reviews that suggestion rather than searching the internet for a dollar P&L figure.

Where the number comes from

Input	Source	Purpose
Latest/current price	Alpaca snapshot or latest daily bar	End-of-period market value proxy
Previous close	Alpaca prior daily bar	Return comparison point
Component weight	Fund configuration in PostgreSQL	Combines multi-component returns
Opening assets	Previous finalized valuation	Dollar base for suggested P&L
Opening units	Settled fund-unit ledger	Denominator for NAV

Calculation

- **Component return** = current price / previous close – 1
- **Fund return** = sum of component weight × component return
- **Suggested daily P&L** = opening assets × fund return
- **Closing assets before flows** = opening assets + daily P&L
- **NAV per unit** = closing assets before flows / opening units

Current production condition

The deployed backend is sending an authenticated Alpaca request with the IEX feed, but Alpaca responds 401 Unauthorized. Therefore the automatic suggestion is unavailable today. A manager_entry value has no internal market-data proof; it must come from a broker/custodian statement or another authorised accounting source and must be explained in the audit note.

What the Manager clicks

1. Choose a managed fund and a date not already present in Finalized valuation history.
2. Click Refresh calculation. If a suggestion appears, review its return, prices, weights, and timestamp.
3. If the service is unavailable, enter only a prepared externally verified P&L and write the source/reason in Audit note.
4. Click Preview calculation. Check opening assets, P&L, closing assets, units, NAV, and investor allocations.
5. Finalize once only when the preview is correct. The record becomes historical and is not casually editable.

4. Verified P&L example

VOO on 4 August 2026

Metric	Verified value	Meaning
Opening assets	\$5,105.00	Fund assets before the day's P&L and flows
Manager-approved daily P&L	+\$10.00	Manual presentation-readiness value; not Alpaca-derived
Closing assets before flows	\$5,115.00	$\$5,105.00 + \10.00
Opening units	5,000.0000	Units outstanding before same-day flows
Finalized NAV	\$1.02300000	$\$5,115.00 / 5,000.0000$
Test Investor ownership	40.00%	$2,000 / 5,000$ opening units
Test Investor allocated P&L	+\$4.00	$40\% \times \$10.00$
QA Investor allocated P&L	+\$0.78	$7.8355\% \times \$10.00$, displayed rounded
External ownership allocation	+\$5.22	Remainder; all allocations sum to \$10.00

What performance changes

Daily P&L changes NAV and each holder's value. It does not create units. Subscriptions create units and redemptions remove units at the finalized NAV; those cash flows are kept separate from investment performance.

Verified same-day settlement arithmetic

Action	Amount	Units at \$1.023 NAV	Result
VOO subscription	\$12.34	+12.0626	Completed by Operations once
VOO redemption	\$7.89	-7.7126	Approved, payout stage, completed
Net	+\$4.45	+4.3500	$2,000.0000 \rightarrow 2,004.3500$ units

Portfolio value moved from \$5,272.00 to \$5,276.45, exactly matching the net cash flow of +\$4.45. Daily P&L remained \$4.00 because cash flows do not masquerade as performance.

5. Admin operator reference

Page / control	What it does	Role handoff / presentation point
Dashboard	Shows access and oversight summary	Start here; Admin observes rather than settles
Users → Edit	Changes full name, email, role, password reset value, active status	Confirm role and status; important changes are audited
Users → Active	Deactivates access without erasing financial history	Prefer deactivation; no destructive delete-user control
Settings → Invite User	Fields: full name, email, role; creates/sends invitation	Hands a new identity to the selected role workflow
Pending invitations → Resend	Resends an existing invitation	Use only for a real prepared address
Pending invitations → Delete	Removes the invitation, not a financial user ledger	Do not confuse invitation deletion with user deletion
Operations invite requests → Approve / Reject	Reviews access requests raised by Operations	Shows two-person access administration
Investor-manager assignment	Maps an Investor to a Manager	Determines which Manager sees/serves that Investor
Fund Flows	Read-only financial oversight	Operations owns settlement
Valuations	Reviews finalized NAV/P&L history	Manager owns finalization
Transactions	Searches and filters recorded activity	Evidence and reconciliation
Audit Logs	Shows actor, role, action, time, and entity details	Close the demo with traceability
Articles	Creates/edits educational content	Visible to Investors without changing holdings
Account security	2FA and account-security controls	Set up before the presentation; do not spend live time here

Adding a user proof

A QA Investor account already exists and has a settled VOO holding. For the live demo, create or show an invitation using a prepared address; do not send an unplanned email to a real person.

6. Investor operator reference

Control	What it actually does	Downstream role / safeguard
Account selector	Chooses the portfolio account being viewed or used	Flows and holdings remain linked to that account
Add Account	Creates another portfolio account/strategy container	Does not add cash or units
Edit strategy	Updates the account strategy label	Does not rebalance broker holdings automatically
Deposit to Fund	Select account, fund, exact amount; creates subscription	Demo PayNow must record the same amount; Operations settles
Demo QR / Simulate payment	Displays a fixed dummy QR and records exact demo receipt	No real money moves; no units yet
Withdraw	Select account, settled fund position, amount	Cannot exceed settled redeemable value; Operations settles
Funds	Search, sort, and filter available fund products	Invest opens the subscription workflow
Fund Flows	Shows request ID, type, amount, payment state, and status	Explains whether Investor, Manager, or Operations is blocking progress
NAV P&L Allocations	Shows finalized daily allocation for this Investor	Derived from Manager-finalized fund P&L and opening ownership
Account Value	Settled holdings valued at latest finalized NAV	Pending requests do not become holdings
Daily / monthly / YTD	Shows performance metrics separated from cash flows	Deposits/withdrawals must not appear as investment return
Performance dates + Calculate	Recalculates the selected analytics period	Use a populated period during the demo
Fund Allocation	Pie chart of settled value by fund product	Uses the same authoritative positions as the table
Recent Fund Activity	Latest subscriptions/redemptions	Replaced unrelated AAPL/MSFT sample transactions
Fund Investments	All settled positions: units, NAV, cost basis, value	Now matches the allocation chart
PDF	Exports portfolio summary	Demonstrate only after holdings load
Email Summary	Emails portfolio summary to the signed-in Investor	External SMTP delivery may be delayed
Articles	Reads educational content by category	No effect on financial records
Feedback	Fields: subject and message; submits service feedback	Operations can respond
Account security	2FA/account-security controls	Preconfigure before the live run

7. Manager operator reference

Page / control	What it actually does	Relationship to other roles
Dashboard / Investors	Shows Investors assigned to this Manager	Admin controls assignments
Funds → Your Funds	Shows only Manager-owned products, not the full catalogue	Operations approval makes new funds investable
Create Fund: name / description	Defines the product identity	Does not create an Alpaca fund or investor position
Strategy / risk	Classifies mandate and risk presentation	Investor sees these catalogue labels
Underlying search	Adds ticker components from market lookup	Defines the simplified market-return model
Allocation weights / Remove	Sets component weights; total should be 100%	Weights drive automatic suggested return
Create	Submits a fund for Operations review	It becomes visible/investable only after approval
Performance dates / Refresh	Loads historical performance attribution	Manager analysis; does not modify holdings
Weight fields / Run what-if	Models a hypothetical allocation	Scenario only; does not trade or finalize NAV
Valuation Fund / Date	Chooses one managed fund and business date	Date must not already be finalized
Refresh calculation	Requests automatic market-data suggestion	Currently blocked in AWS by Alpaca 401
Daily fund P&L	Prefilled suggestion or externally verified manual fallback	Changes NAV/value, never unit count
Audit note	Records source/reason for manual value or adjustment	Admin/Operations audit evidence
Preview calculation	Shows assets, units, NAV and ownership allocation before committing	Required review point before Operations settles
Finalize valuation	Writes one finalized daily valuation	Unlocks applicable Operations settlement
Finalized history	Shows past P&L, NAV, date, source, actor	Evidence for Investor allocations and audit
Transactions / search / side / CSV	Inspects activity and exports reconciliation data	Read/analysis function, not settlement
Articles / security	Educational content and account protection	No financial authority

8. Operations operator reference

Control	What it actually does	Required check / handoff
Fund Flows search	Searches email, name, or request ID	Use the Investor's unique reference
Type / status filters	Narrows subscription, redemption, and lifecycle state	Use Pending Ops Team during the demo
Requested / Received	Shows fixed requested amount and recorded receipt	Must match before a demo subscription settles
Verify & Complete	One-step demo PayNow subscription settlement	Issues units once at finalized NAV; action disappears after completion
Approve redemption	Authorises the request and moves it toward outgoing transfer	Does not yet remove units
Complete redemption	Confirms payout was sent and redeems units	Use only after outgoing transfer; action is one-time
Start / Retry Payout	Provider payout action when Stripe Connect is used	Not needed for manual/demo path
Reject	Rejects a pending request	A reason should explain the decision in notes/audit
Settlement status message	Explains missing valuation/payment/payout condition	Read it instead of repeatedly clicking
Fund Reviews → Approve / Reject	Reviews Manager-created fund products	Approval makes eligible products investable
Audit Logs	Reviews sensitive actions	Confirms actor, role, time, entity
Feedback → Respond	Closes Investor feedback loop	Service task, not financial settlement
Invite Requests	Fields: full name, email, requested role	Admin independently approves/rejects
Account security	2FA/account protection	Set up before presenting

Why withdrawal has two Operations actions

Approval confirms the redemption is authorised. Completion confirms the outgoing payout has actually been sent and only then removes units. Demo PayNow subscriptions use one Verify & Complete action because receipt is already recorded before Operations review.

9. Product suite and prepared Investor

Catalogue truth

- The original seed had 20 rows: 13 fund products and 7 ordinary stocks.
- The seven stock rows were removed from the Investor fund catalogue.
- The valid demonstration suite is 13 products: QQQ, VOO, VTI, SPY, BND, AGG, VYM, SCHD, SOXL, TQQQ, TLT, SCHR, and VIG.
- Manager 'Your Funds (3)' is a manager-ownership count, not the total catalogue count.

investor@fundinv.com now owns 10 settled products

Established	Added through normal demo workflow
QQQ — Invesco QQQ Trust	TLT — iShares 20+ Year Treasury Bond
VOO — Vanguard S&P 500 ETF	AGG — iShares Core US Aggregate Bond
	SCHR — Schwab Short-Term US Treasury ETF
	SCHD — Schwab US Dividend Equity ETF
	VIG — Vanguard Dividend Appreciation ETF
	VYM — Vanguard High Dividend Yield
	BND — Vanguard Total Bond Market
	VTI — Vanguard Total Stock Market

How this was achieved

Eight \$25 subscriptions were created through the Investor fund catalogue, paid with the demo PayNow action, and completed by Operations. No direct database balance edit was used. SPY has a paid request but remains pending because its Manager valuation/mapping condition is not satisfied; SOXL and TQQQ are not part of this Investor's prepared holdings.

10. Presentation run of show

12–15 minutes, business-first

Time	Role / screen	Action	What to say
0:00	Opening	State the four-role control model	Access, valuation, settlement, and reporting are separated
0:45	Admin	Show prepared invitation or create one	Admin grants roles but cannot settle money
2:00	Investor / Funds	Search VOO; open Invest	Investor chooses a fund product and exact amount
3:00	Investor / QR	Submit and simulate demo PayNow	Amount/reference are locked; no units yet
5:00	Manager / Valuation	Refresh, explain source, preview, show finalized history	NAV changes unit value, not unit count
8:00	Operations / Flows	Match request and receipt; Verify & Complete	Independent settlement; one-time action
10:00	Investor / Portfolio	Show activity, positions, P&L, chart	Cash flow and performance remain separate
11:30	Investor / Withdraw	Submit safe redemption	Cannot exceed settled position
12:30	Operations	Approve then complete after payout step	Two stages protect outgoing money
14:00	Admin / Audit	Show audit evidence and close	Every sensitive step has accountable ownership

Pre-login setup

- Use four separate browser profiles or a mix of normal/private windows. Tabs in one browser profile share the same authentication cookie and will overwrite each other's role session.
- Admin: admin@fundinv.com / admin123
- Manager: manager@fundinv.com / admin123
- Operations: operations@fundinv.com / admin123
- Investor: investor@fundinv.com / investor123
- Use localhost only for local testing; use the CloudFront URL for the hosted presentation.

11. Seamlessness and remaining limitations

Overall assessment: strong classroom demo, not defect-free production accounting

The role handoffs and core accounting controls are coherent and usable. The demo is smooth when role windows and a finalized NAV are prepared. It is not fully seamless because external market-data authentication and CI/CD credentials still need repair.

Issue	Presentation impact	Recommended handling
Alpaca returns 401 in AWS	Refresh calculation cannot prefill suggested P&L	Use prepared audited manual P&L; repair AWS credential later
GitHub Actions AWS credentials expired/invalid	Push does not auto-deploy	Current release was manually deployed; repair OIDC/static credential setup after presentation
Single-instance ASG desired capacity	Good for small demo, not highly available	Do not claim enterprise HA; load test before scale claims
SPY paid request pending	Visible blocker if selected	Use VOO or QQQ for the live workflow
Manual manager_entry provenance	System trusts external Manager source	Use audit note and authoritative statement; never Google dollar P&L
Simplified market-return valuation	Does not include exact broker quantities, cash, fees, dividends, liabilities	Describe as a prototype valuation model with Manager accounting adjustment
Shared Alpaca holdings concept	Could be mistaken for Investor-specific holdings	Underlying holdings section is hidden/redesigned; use Fund Investments for Investor ownership

12. Final preflight and recovery checklist

Ten minutes before presenting

- Open the CloudFront portal in four isolated browser profiles and confirm each role is still signed in.
- Keep VOO or QQQ selected; do not use SPY for the transaction demo.
- Confirm the valuation date is not already finalized before promising a live finalization.
- Prepare a small, memorable subscription and redemption amount.
- Keep paynow_demo; do not use real money or Stripe payout actions.
- Have this guide open on a second screen and keep credentials out of the projected browser history.
- Do not edit AWS secrets, email settings, roles, or fund mappings during the presentation.

If something stops

Symptom	Meaning	Safe response
Date already finalized	Final records are protected	Choose a truly unfinalized date or show history
Market suggestion unavailable	External market service failed/auth failed	Use prepared verified figure with audit note; state limitation
Operations button disabled	Payment, valuation, fund, or state prerequisite is missing	Read the status message; do not repeatedly click
Investor values look stale	Previous role mutation has not been reloaded	Refresh only after Manager/Operations completes
Invitation email delayed	Notification provider delay	Show invitation record; access control remains in the database
Role login replaces another tab	Same browser cookie shared	Use separate profiles/private windows

Closing line

FundInv demonstrates a complete controlled lifecycle: Admin grants access, the Investor requests a transaction, the Manager values the fund, Operations independently settles money and units, and the Investor receives transparent holdings and performance reporting.